Context Engineering

Language Models

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This Part: Context Engineering

- in-context learning
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In-Context Learning (ICL)

the model's ability to condition its outputs on patterns, instructions, or demonstrations present in the context window, *without updating parameters* (in contrast to fine-tuning)

→ not really learning (rather inference-time pattern recognition: locating and using relevant learnings from training)

ICL is the emergent phenomenon that makes prompting powerful.

Few-Shot ICL

zero-shot ICL:

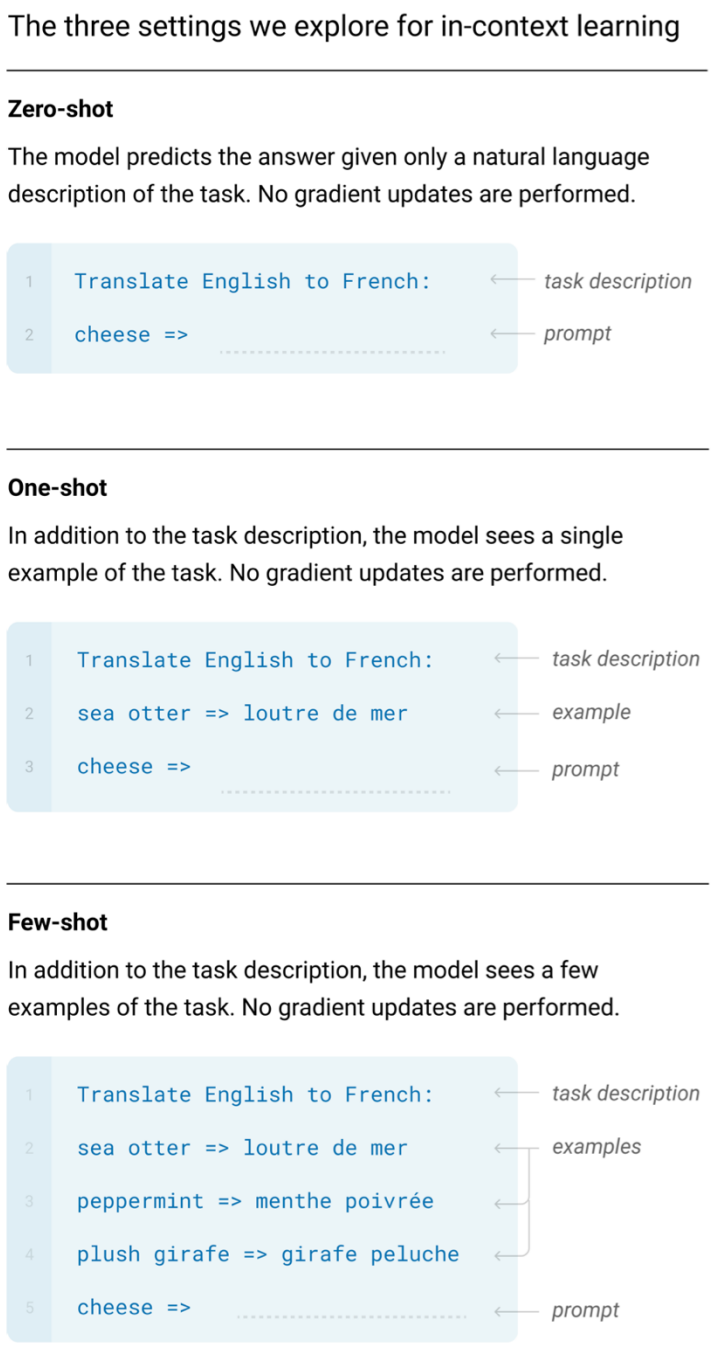
no examples, just instructions

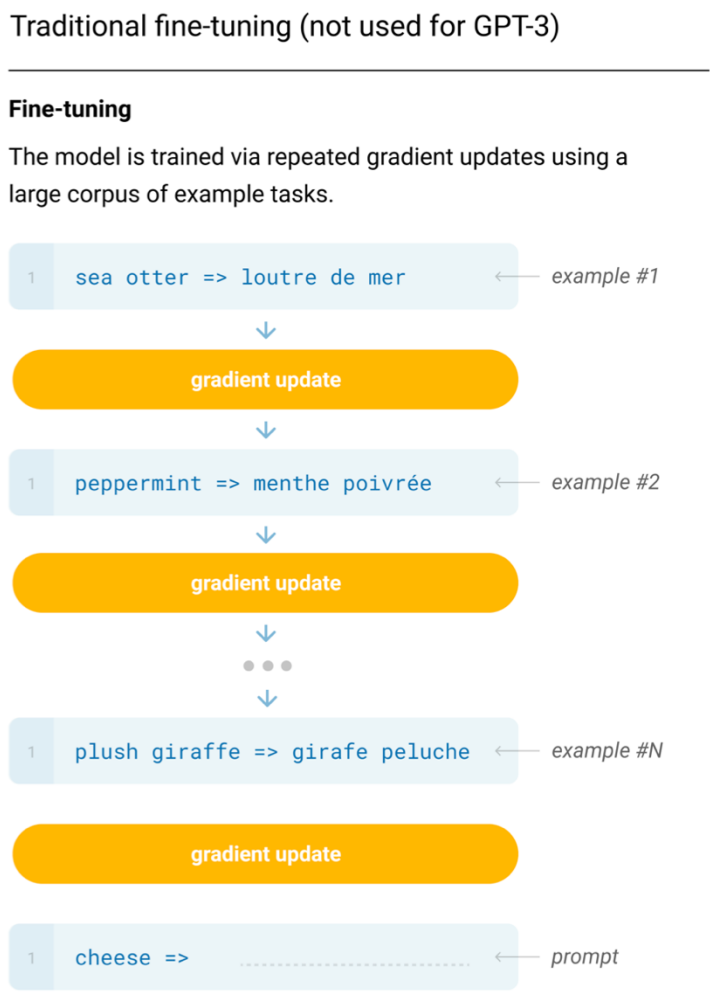
→ instruction following: model infers task purely from natural-language description

few-shot ICL:

conditioning on a few input-output examples

→ demonstration-based: model “learns“ from examples





GPT-3

large context length also allows many-shot ICL

Prompt Engineering

prompt: user interface to the model

prompt engineering: shape model behavior using well-structured prompts

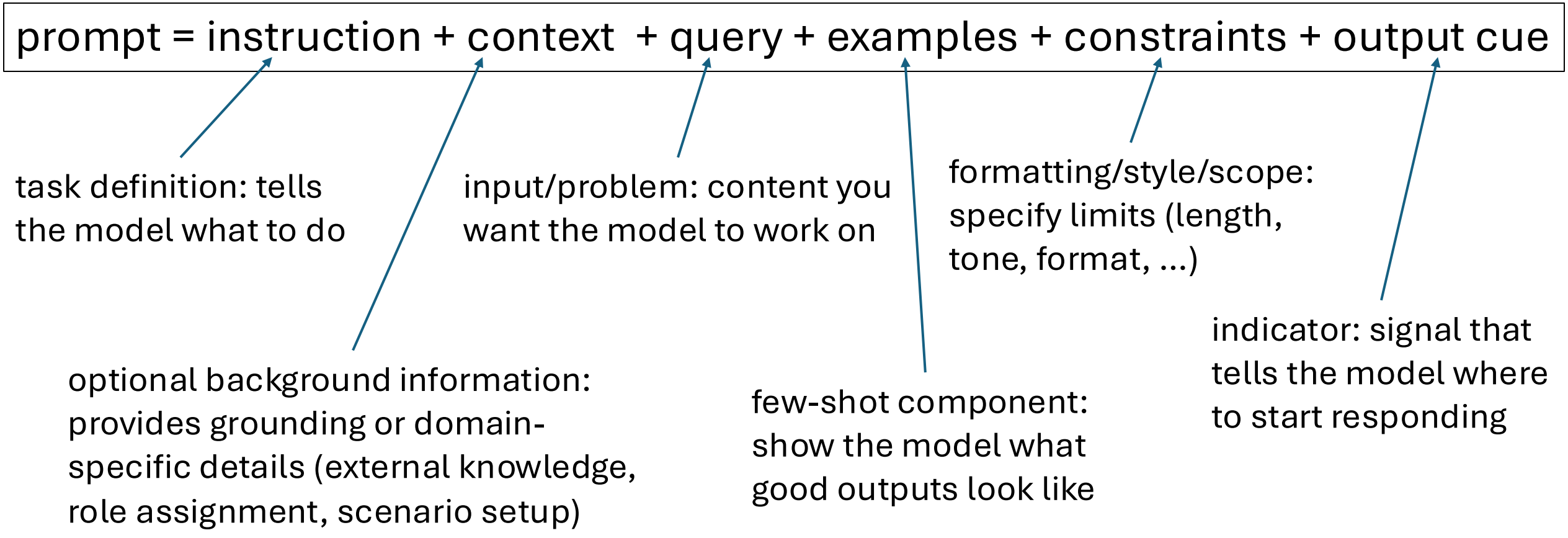
difference from traditional user interfaces:

probabilistic, not deterministic (shape the *distribution* of the model's responses)

→ good prompts can improve a model's performance and reduce hallucination, bias, or inefficiency

prompt engineering, GPT guide

Typical Prompt Structure



Example Prompt

Summarize the following text in one sentence.
Text: "Large language models show emergent abilities when scaled."
Answer:

- instruction: *Summarize the following text in one sentence.*
- query (input/problem): *Large language models show emergent behavior when scaled.*
- constraint: implicit → *in one sentence*
- output cue: *Answer:*

Simplest Possible Prompt

often query phrased as instruction:

> Summarize: Large language models show emergent abilities when scaled.

but without instruction model just completes/continues the query text (usually not really useful):

> Large language models show emergent abilities when scaled.

At the bare minimum, you only need the query. But for reliable, controllable outputs, you usually need to scaffold it with instruction, context, constraints, etc.

Best Practices: Clear Instruction & Query

Tell me about photosynthesis.

better:

Explain photosynthesis in simple terms suitable for a 12-year-old, using a short paragraph and one analogy.

→ clearly state what you want the model to do (best when specific, unambiguous, and outcome-focused)

Best Practices: Give Context

Write about the causes of the French Revolution.

better:

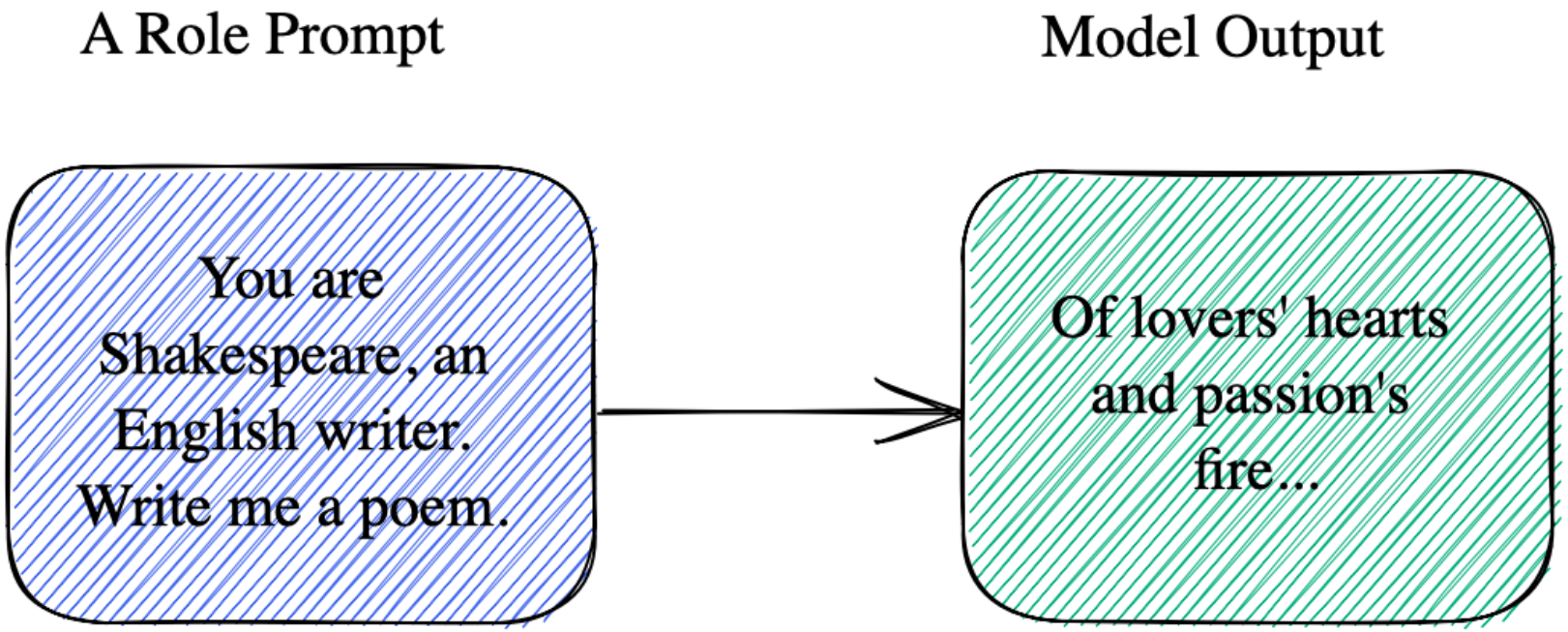
You are a historian writing a study guide for undergraduate students.

Write a concise explanation of the causes of the French Revolution, highlighting 3 major factors.

context: *You are a historian writing a study guide for undergraduate students*.

→ role assignment and audience specification steer tone and depth

Aka Role Prompting



Best Practices: Specify Constraints

List some facts about Mars.

better (if you want specific outputs):

List exactly 3 scientifically verified facts about Mars.

Output them in JSON format with keys: "fact1", "fact2", "fact3".

Output Indicators

earlier models (pre-ChatGPT) were very sensitive to formatting (training datasets contained clear separators)

→ certain cues like *Answer:*, *A:*, *Response:*, or even ### dramatically affected performance (better alignment with training data)

not strictly required anymore for modern instruction-tuned LLMs

but can still be useful for automation or complex prompts (improve consistency, reduce ambiguity):

- classification: `Label:`
- code generation: `Code:`

Examples/Demonstrations

Convert these temperatures from Celsius to Fahrenheit:

25, 30, 35

few-shot examples help for unusual, complex, or under-specified tasks:

Convert these temperatures from Celsius to Fahrenheit:

C: 0 → F: 32

C: 10 → F: 50

C: 20 → F: 68

Now continue for:

C: 25, 30, 35

Important Subtlety: Formatting & Abstraction

minimum effect of examples: enforce formatting patterns (small models)

C: 25 → F: 666

maximum: approximate transformation rule behind examples (large models)

C: 25 → F: 77

But not even the largest LLMs actually derive $F = C \cdot 1.8 + 32$.

instead: interpolation from similar training cases memorized in model weights (not directly interpolating the numerical values, but its vector representations)

→ looks like reasoning, but is just probabilistic pattern completion

→ no algorithmic guarantee (typically breaks for unusual inputs)

modern reasoning models: tool use to the rescue

Structured Design

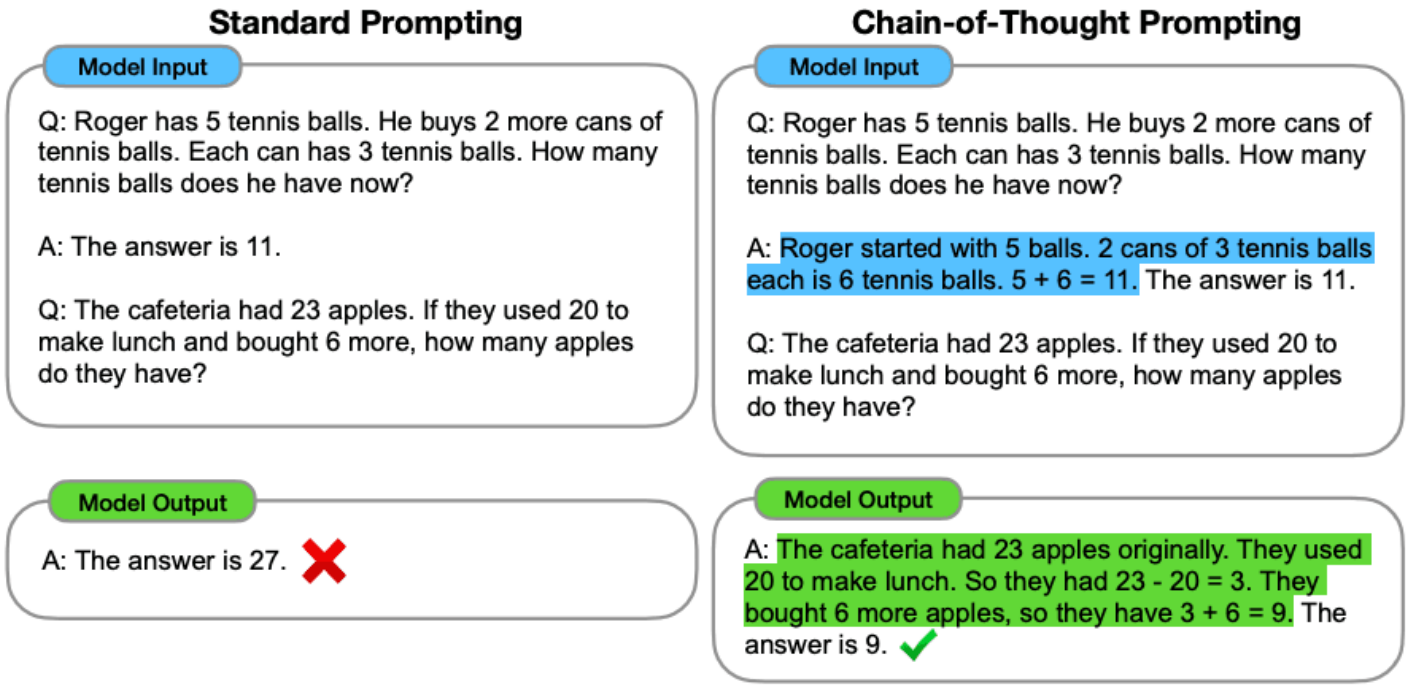
So, prompts aren’t random — they have designable components.

different tasks emphasize different sections:

- creative writing → role/context
- (approximate) reasoning → examples
- structured output → constraints

Chain-of-Thought Prompting (CoT)

explicitly ask the model to **reason step by step** before giving a final answer originally, by providing at least one reasoning example:

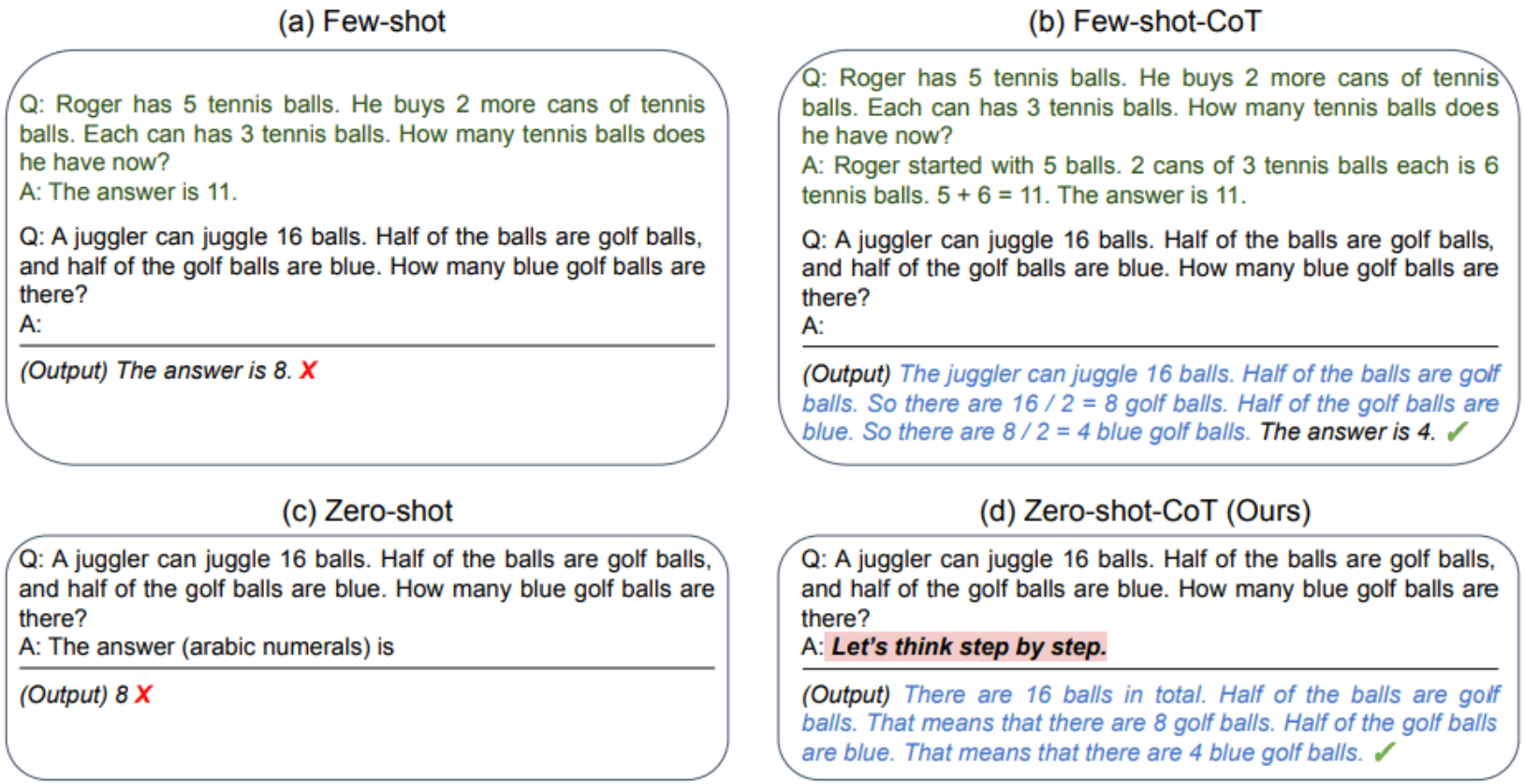


makes model outputs more legible

but no genuine interpretability of underlying LLM reasoning processes

Zero-Shot CoT

also works without examples, just through clever instructions ...



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Hallucinations

knowledge encoded implicitly in weights rather than as explicit facts

→ probabilistic pattern completion can produce hallucinated facts

Instruction: What are Thomas Edison's main contributions to science and technology?

Response: Thomas Edison developed the first practical telephone and invented the light bulb.

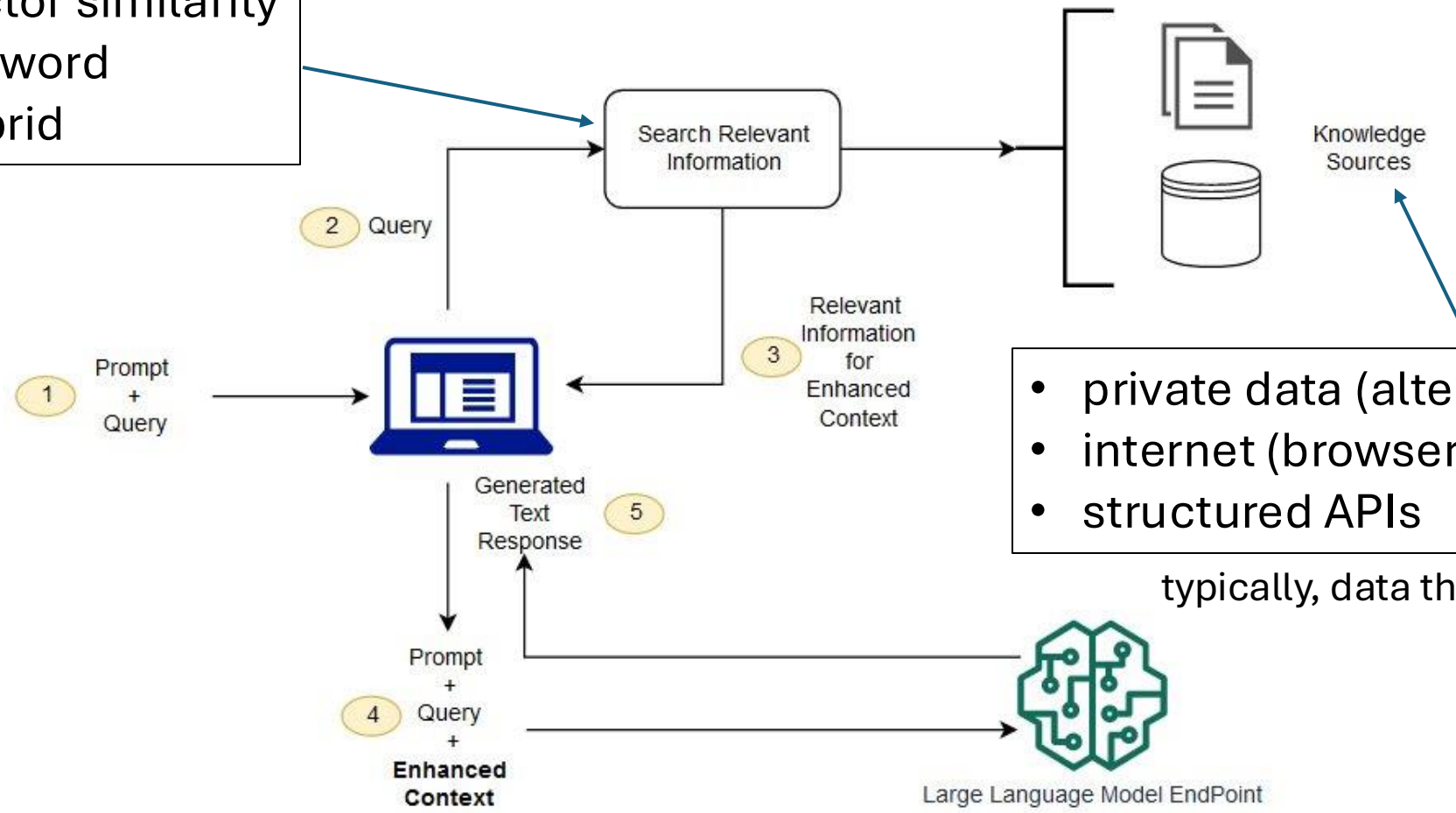
The response is factually incorrect. In reality, **Edison improved the light bulb, building on earlier designs by others**, and **Alexander Graham Bell invented the telephone**.

can be mitigated by retrieving the corresponding information from an external knowledge base and pasting it in context

(prompt engineering → context engineering)

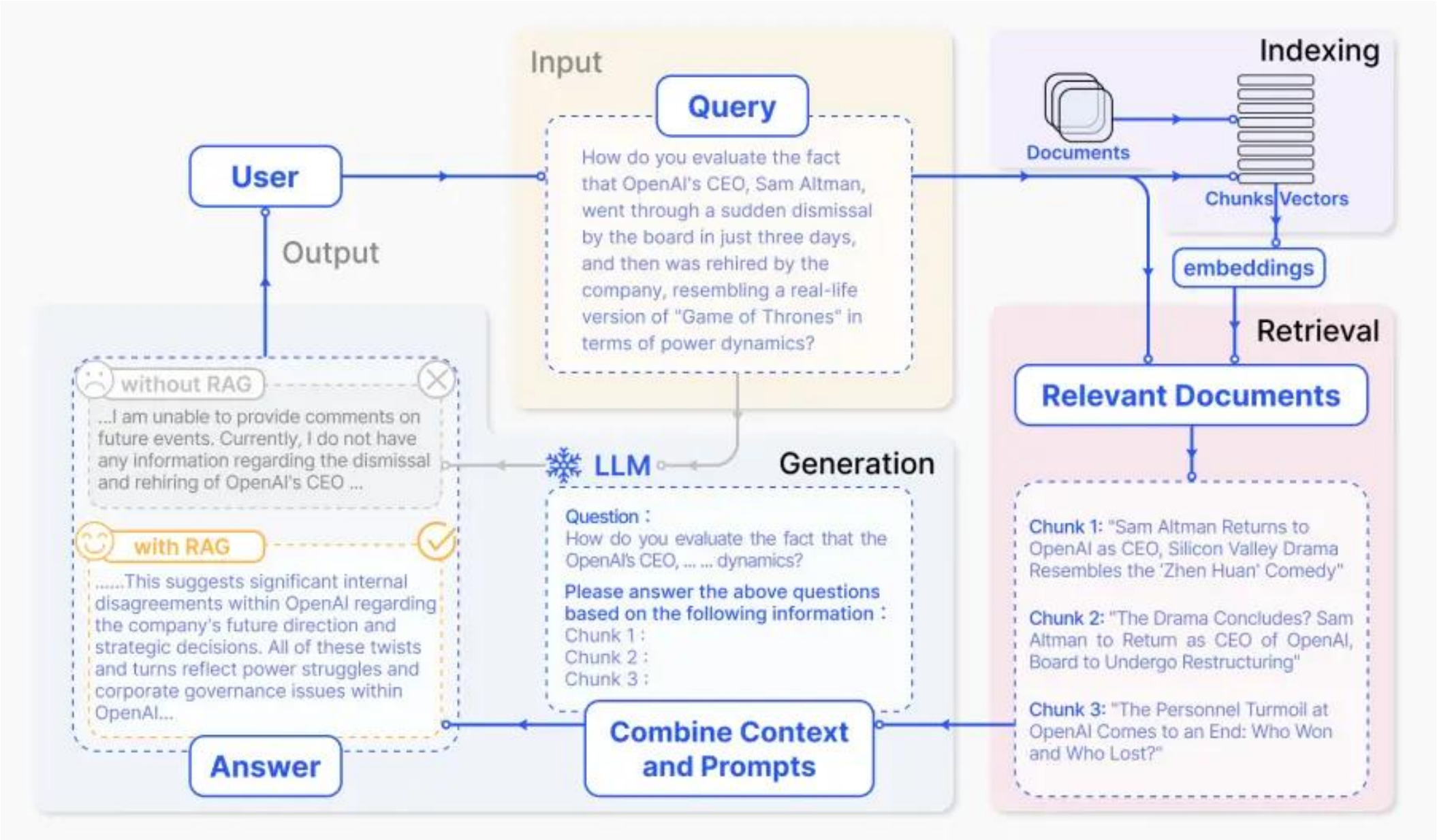
Retrieval Augmented Generation (RAG)

- vector similarity
- keyword
- hybrid



- private data (alternative to finetuning)
- internet (browser as tool)
- structured APIs

typically, data the model wasn‘t trained on


Input
Query
How do you evaluate the fact that OpenAI's CEO, Sam Altman, went through a sudden dismissal by the board in just three days, and then was rehired by the company, resembling a real-life version of "Game of Thrones" in terms of power dynamics?
User
Output
Indexing
Documents
Chunks Vectors
embeddings
Retrieval
Relevant Documents
Chunk 1: "Sam Altman Returns to OpenAI as CEO, Silicon Valley Drama Resembles the 'Zhen Huan' Comedy"
Chunk 2: "The Drama Concludes? Sam Altman to Return as CEO of OpenAI, Board to Undergo Restructuring"
Chunk 3: "The Personnel Turmoil at OpenAI Comes to an End: Who Won and Who Lost?"
LLM
Generation
Question：
How do you evaluate the fact that the OpenAI's CEO, ... dynamics?
Please answer the above questions based on the following information：
Chunk 1：
Chunk 2：
Chunk 3：
Combine Context and Prompts
without RAG
...I am unable to provide comments on future events. Currently, I do not have any information regarding the dismissal and rehiring of OpenAI's CEO ...
with RAG
.....This suggests significant internal disagreements within OpenAI regarding the company's future direction and strategic decisions. All of these twists and turns reflect power struggles and corporate governance issues within OpenAI...
Answer

source

Advanced Prompting Techniques

- self-consistency: aggregation of multiple sampled CoT reasoning paths
- meta-prompts / prompt templates: using one prompt to generate another
- prompt chaining: e.g., 1: extract facts → 2: summarize → 3: critique
- scaffolding: multi-turn prompt orchestration, planning, subtask decomposition
- reflexion: model revisits its output, identifies errors or gaps, and revises
- role-playing and system prompts: e.g., teacher, critic, planner
- automatic prompt generation and optimization (e.g., OPRO)